



AE4238: Aero Engine Technology

Engine Inlets-V: Inlet Distortion

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Faculty of Aerospace Engineering

Y shaped intake



Light combat aircraft (LCA)



Y shaped intake

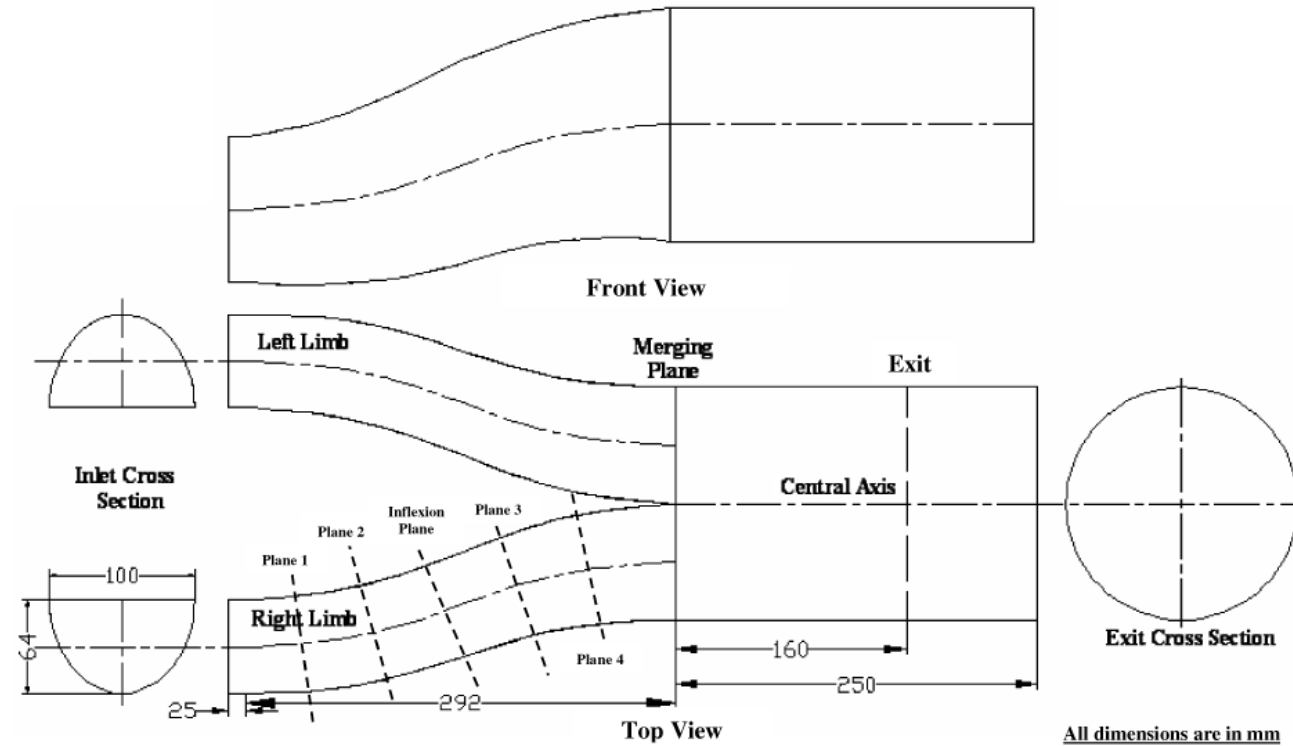


Fig. 3 Schematic layout of the Y duct showing double offsets.

R. K. Singh, S. N. Singh, and V. Seshadri, Journal of Aircraft, Vol 45, 2008

Y shaped Intake

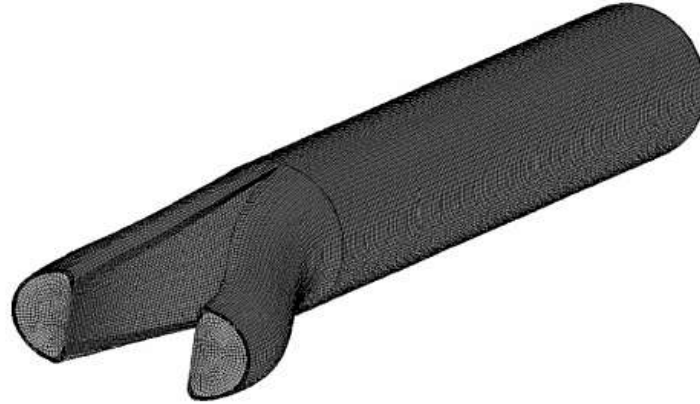
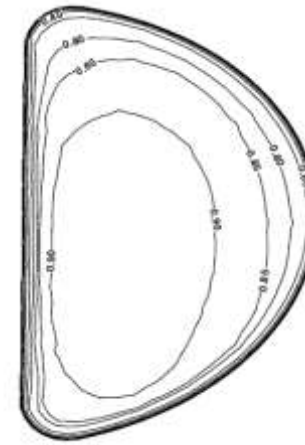
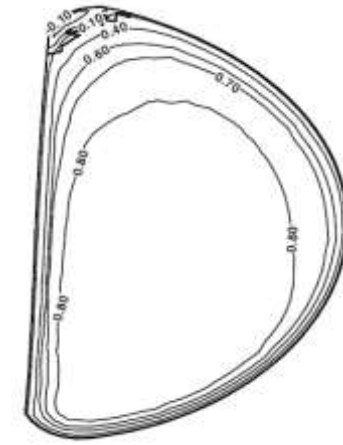


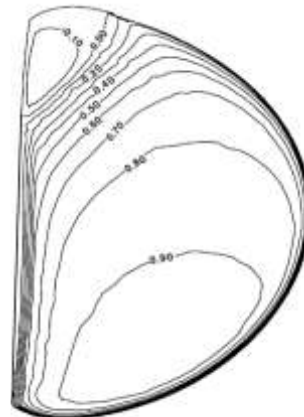
Fig. 4 Discretized 3-D geometry of one of the double-offset Y ducts.



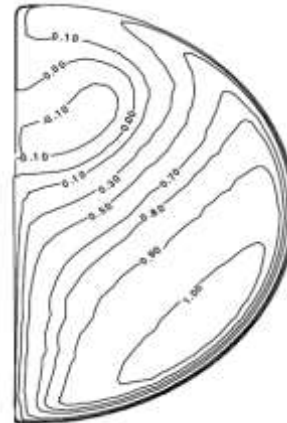
a) Plane 1



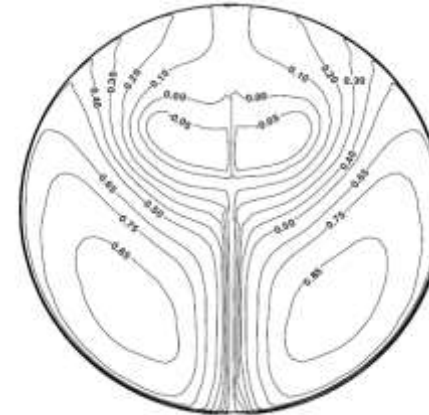
b) Plane 2



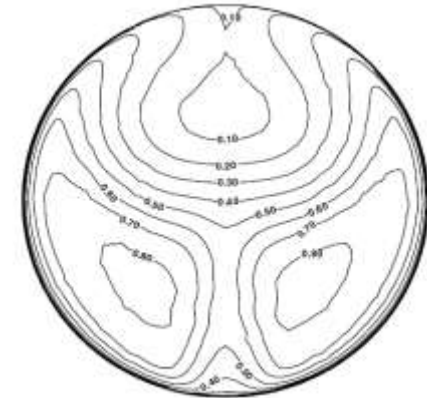
c) Inflexion plane



d) Plane 4



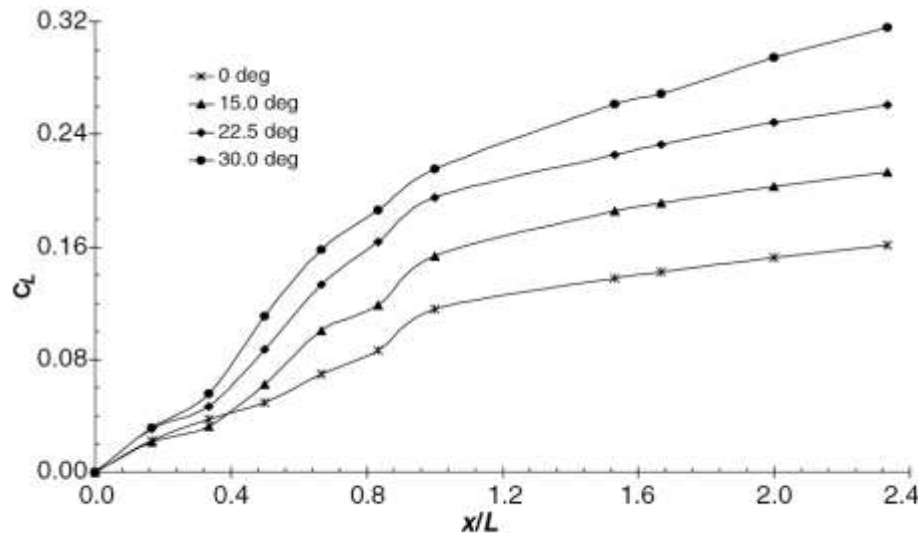
f) Merging plane



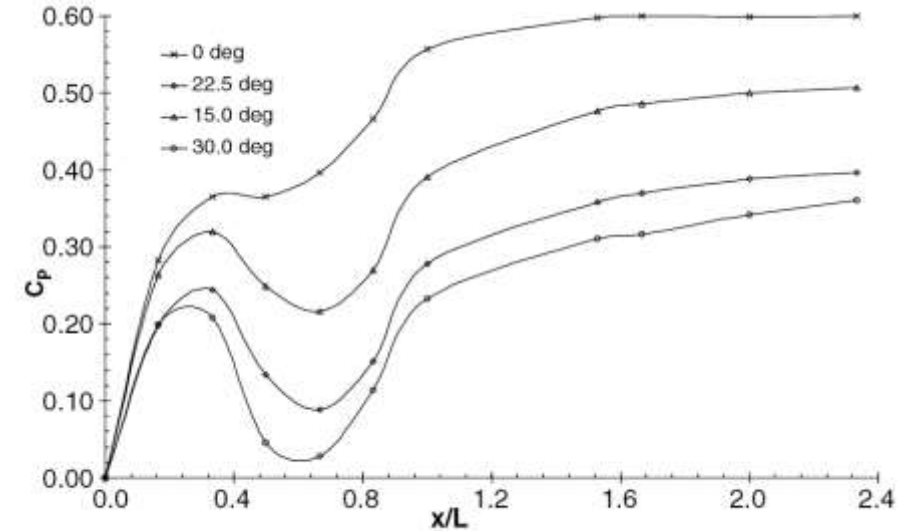
g) Outlet plane

Y shaped Intake

$$C_P = \frac{p_{si} - p_{so}}{\frac{1}{2} \rho U_{in}^2}$$



b) Variation of total pressure loss coefficient



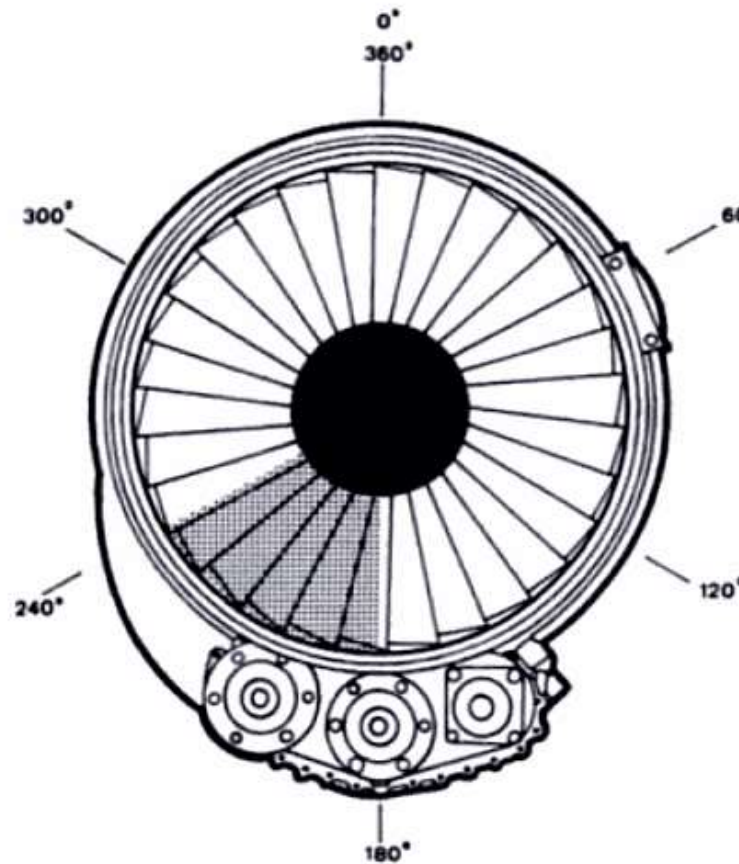
a) Variation of coefficient of pressure recovery



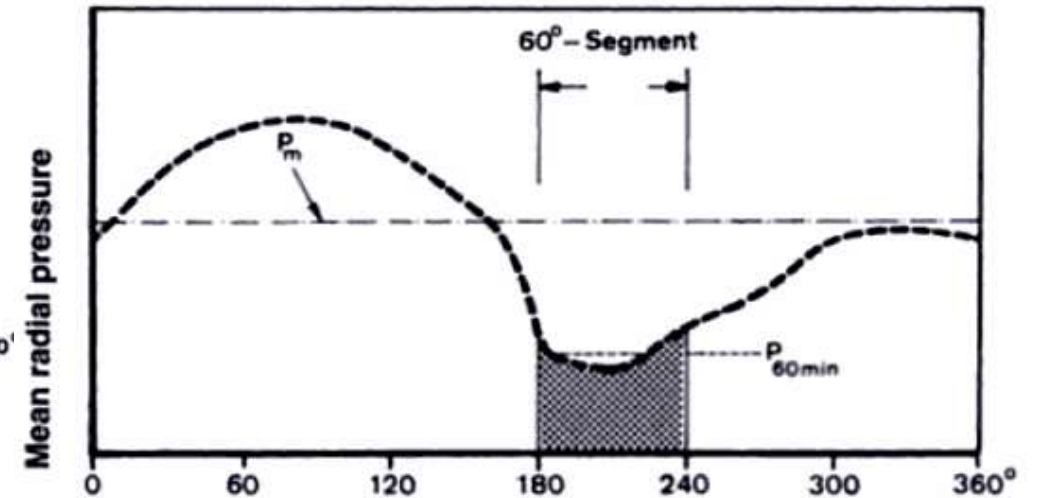
$$C_L = \frac{p_{Ti} - p_{To}}{\frac{1}{2} \rho U_{in}^2}$$

Distortion Parameter

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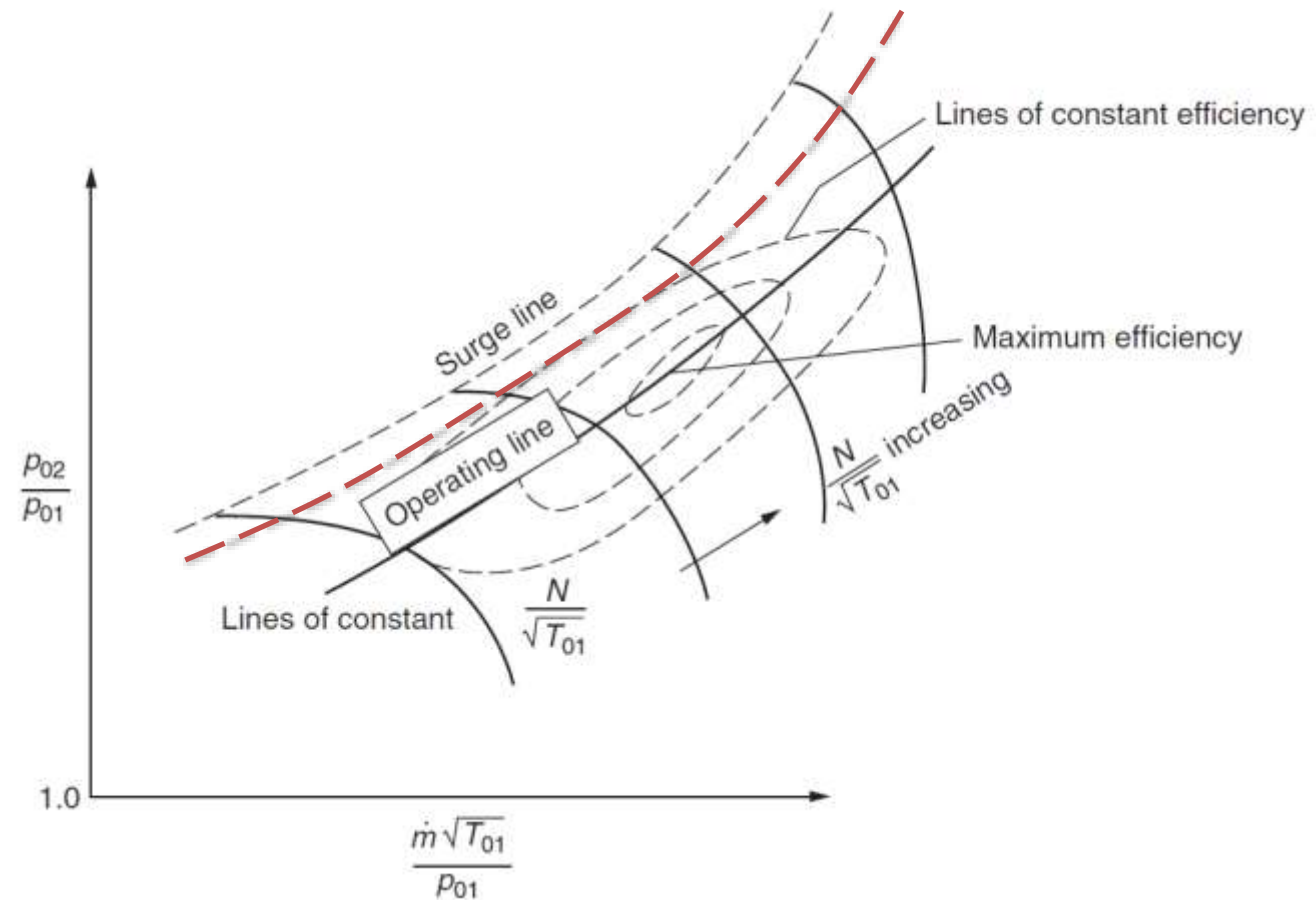
$$\text{Distortion parameter DC60} = \frac{P_m - P_{60\min}}{q}$$



i Explaining distortion factor

Compressor / Fan Map

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Fan Test Rig

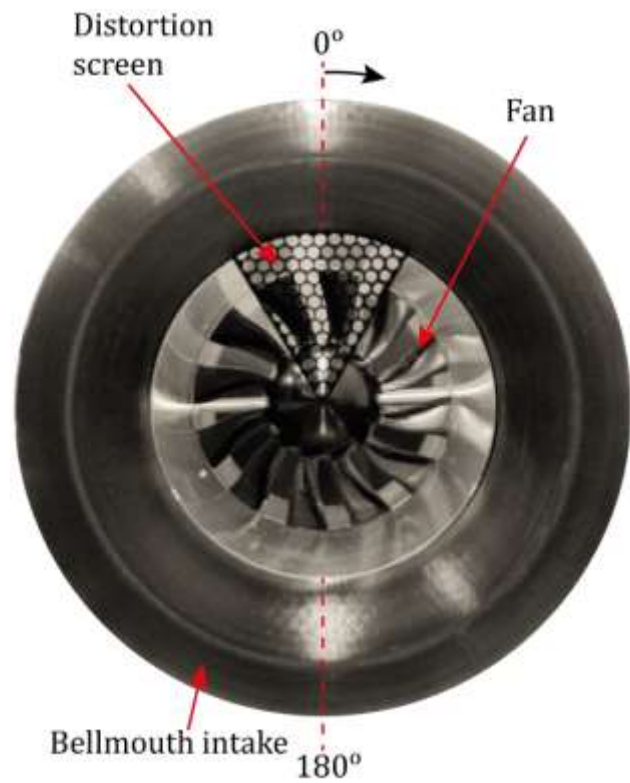
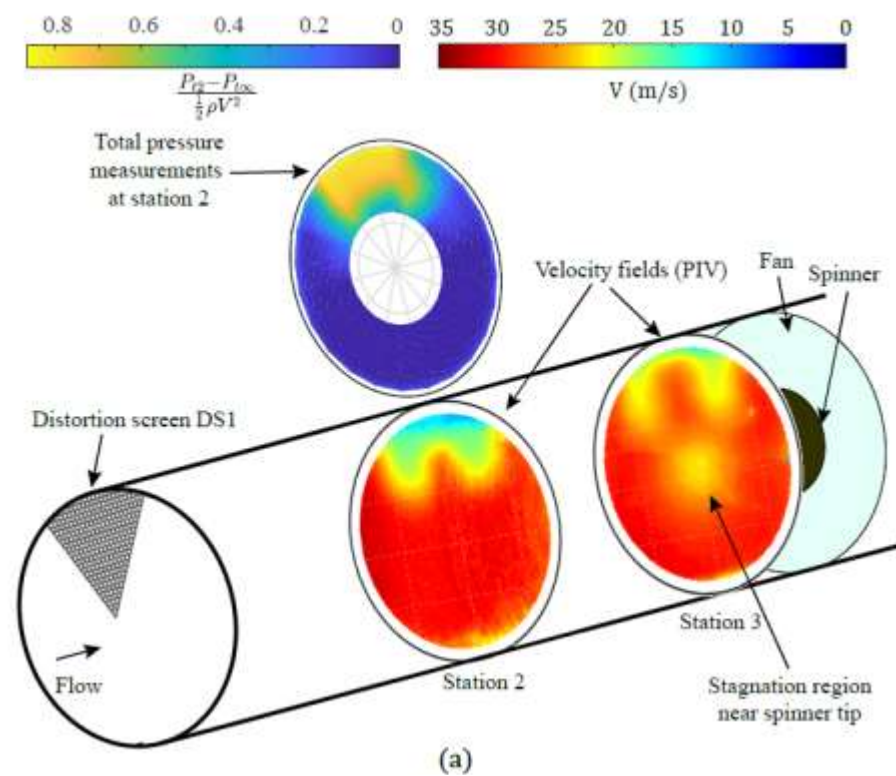


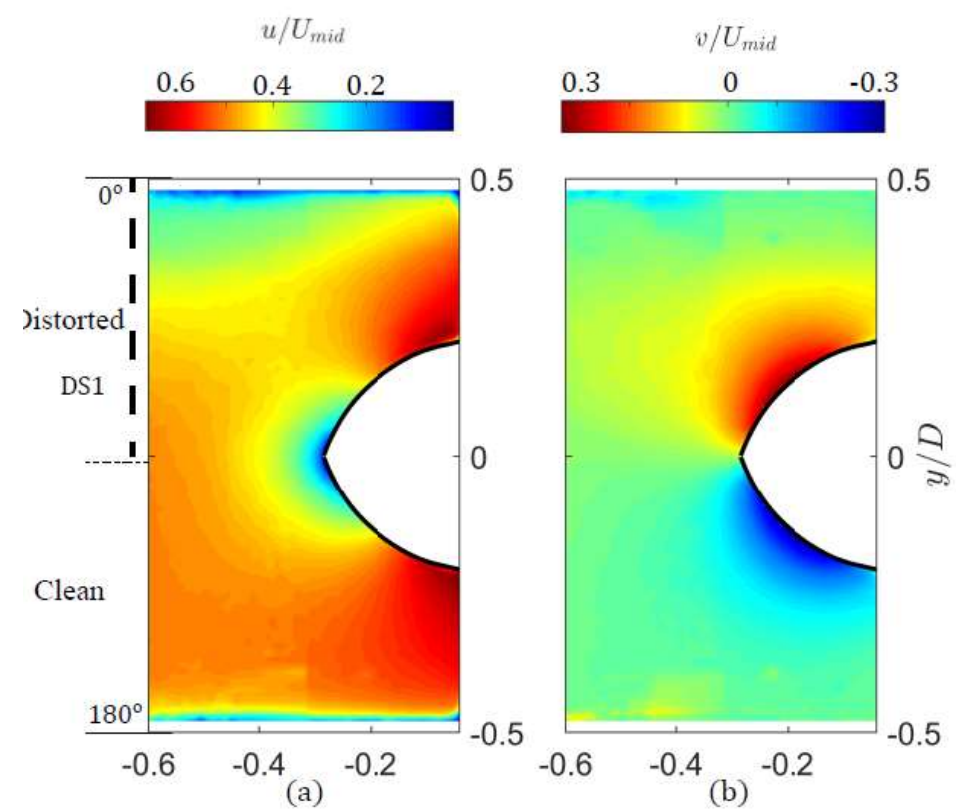
FIGURE 2. Front view of the test rig



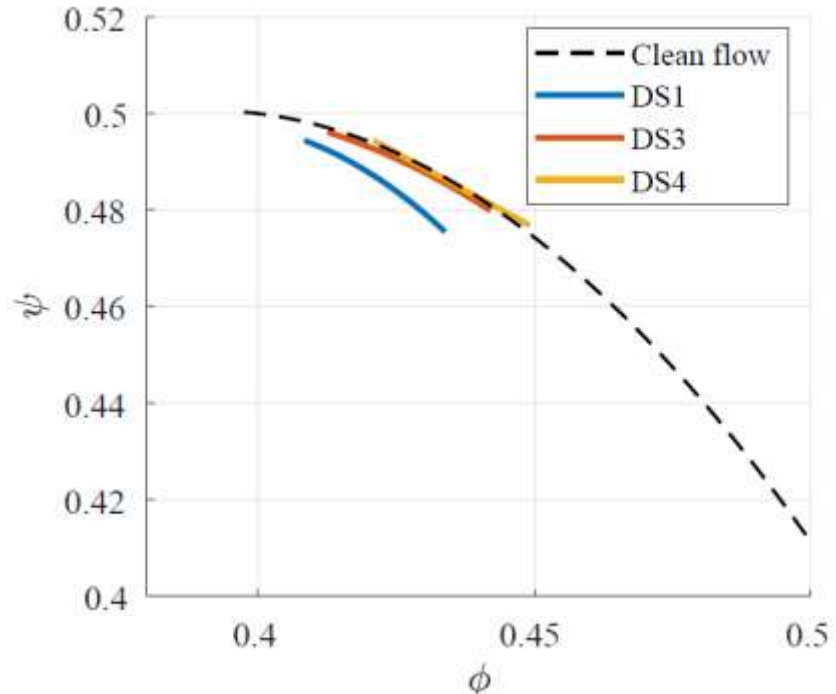
Experimental flow field measurements using Particle Image Velocimetry

S. Tambe, U. B. Oseguera and A. Gangoli Rao "Performance of a low speed axial fan under distortion: An Experimental Investigation" ASME TurboExpo 2020, London.

Distortion screen



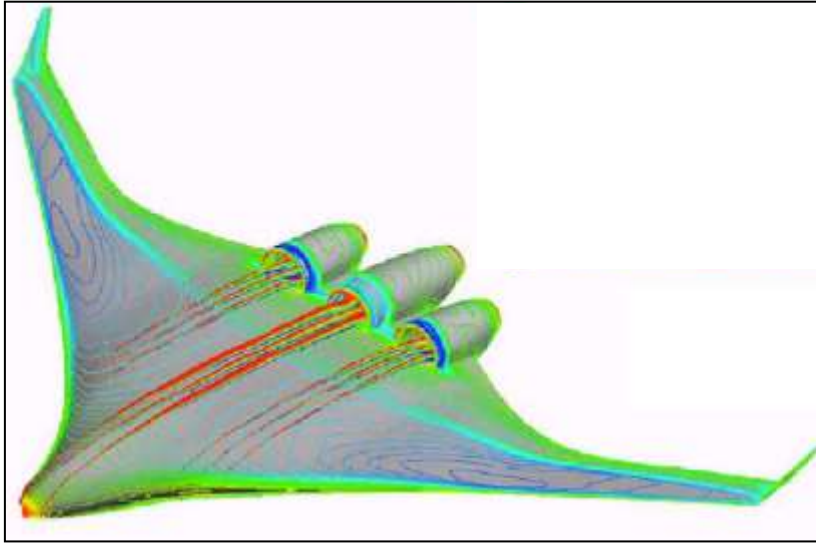
Variation in the axial and radial velocity from PIV measurements



Effect of distortion on compressor performance

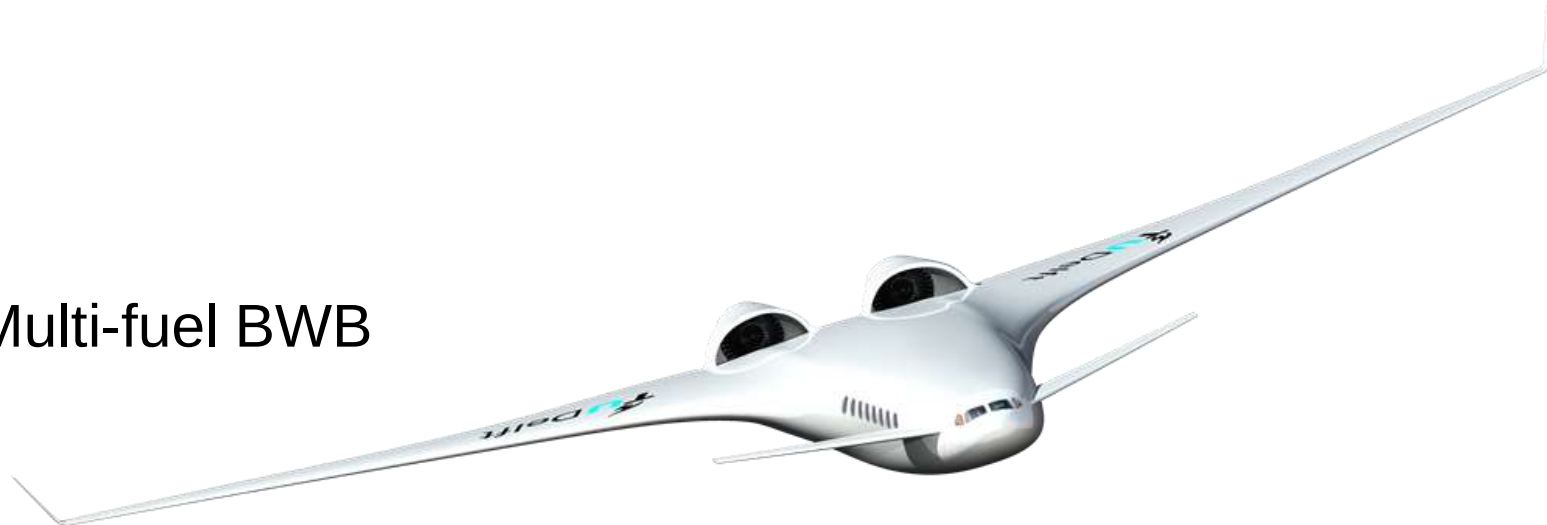
S. Tambe, U. B. Oseguera and A. Gangoli Rao "Performance of a low speed axial fan under distortion: An Experimental Investigation" ASME TurboExpo 2020, London.

Boundary Layer Ingestion

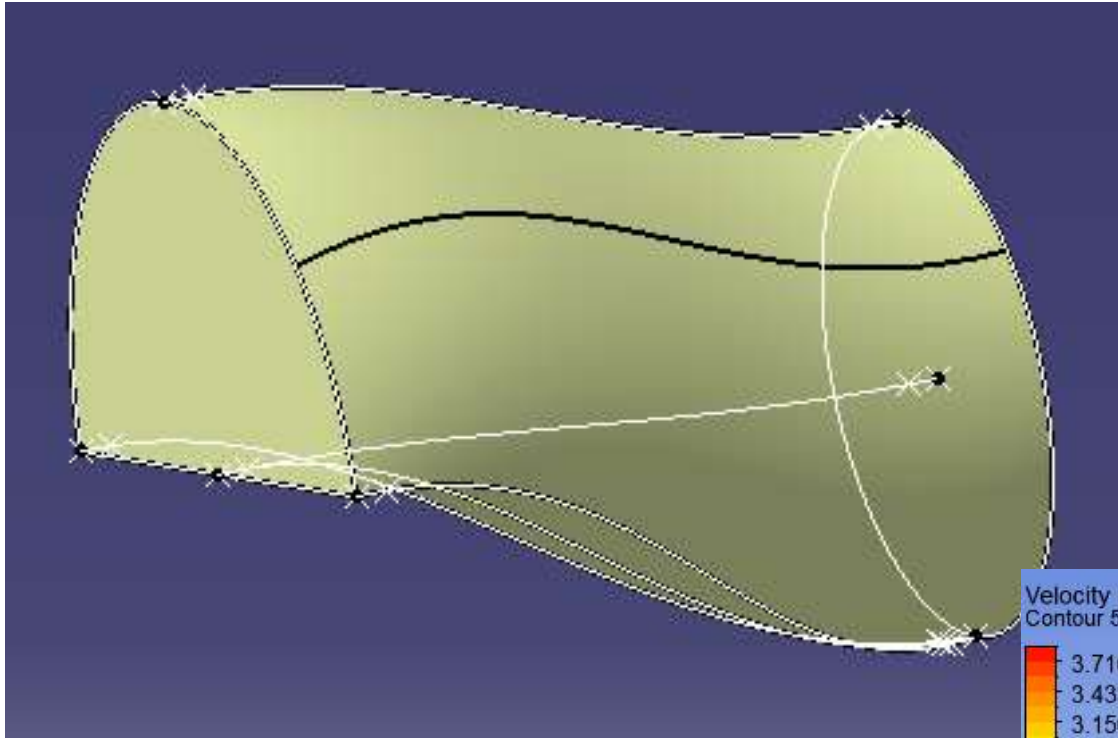


NASA Blended wing body aircraft

AHEAD Multi-fuel BWB



Inlet distortion

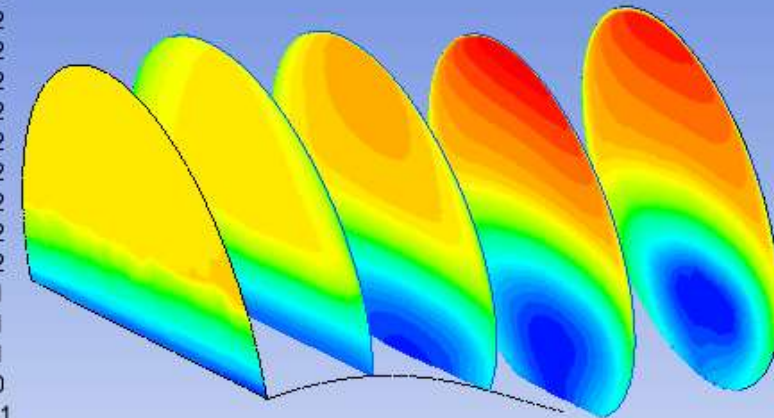


$H = 0.3\text{m}$
 $L = 4.85\text{m}$
 $AR = 1.75$
 $PR = 97.7\%$
 $DC_{60} = 28.45\%$

Velocity u
Contour 5

3.716e+002
3.433e+002
3.150e+002
2.867e+002
2.584e+002
2.301e+002
2.018e+002
1.736e+002
1.453e+002
1.170e+002
8.871e+001
6.042e+001
3.214e+001
3.855e+000
-2.443e+001

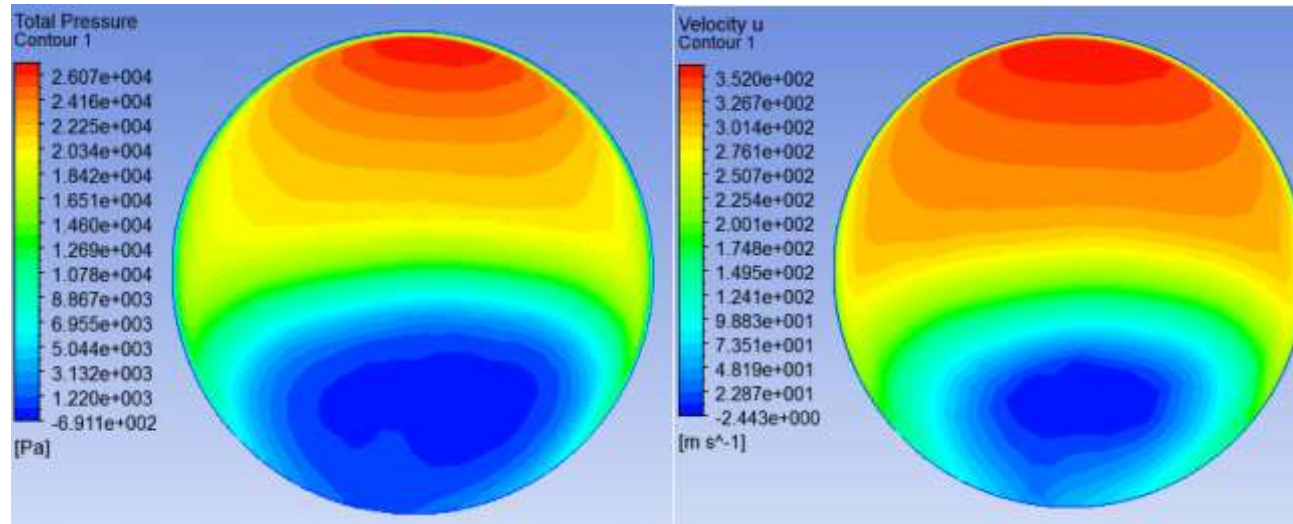
[m s⁻¹]



A. Gangoli Rao, A. Sharma, R. van Dijk, "A CFD Based Parametric Analysis of S-shaped Inlet for a Novel Blended Wing Body Aircraft", 2017

Variation in Inlet Length

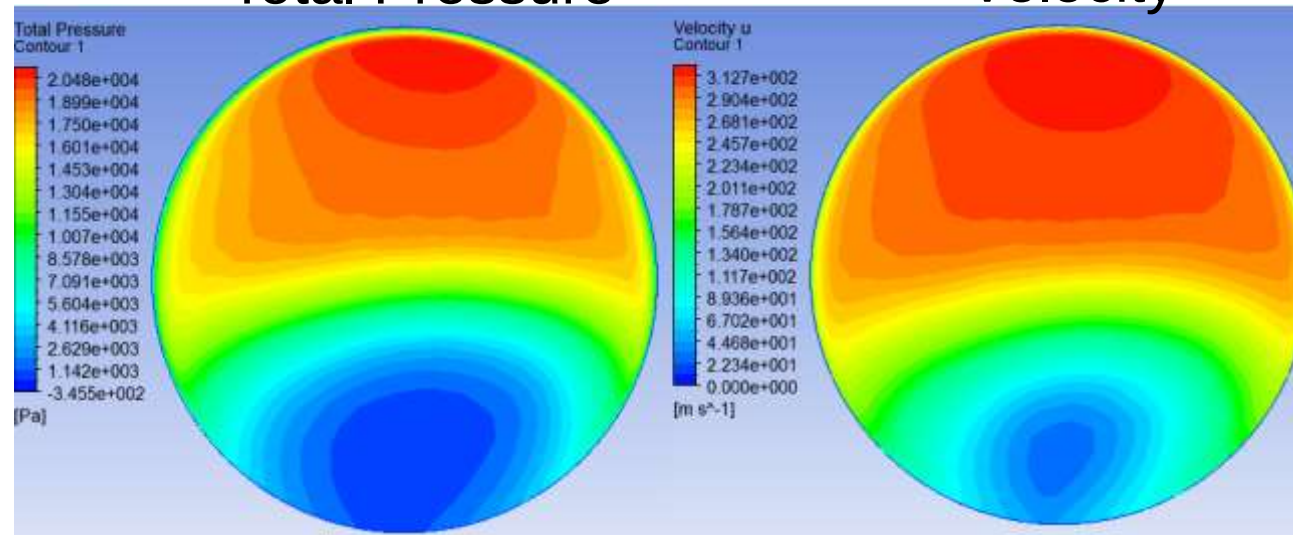
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$L=4.85\text{m}$

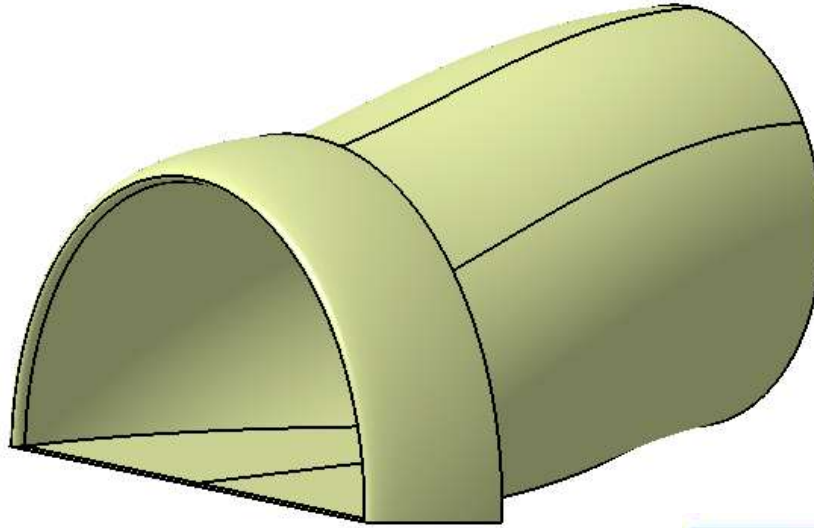
Total Pressure

Velocity



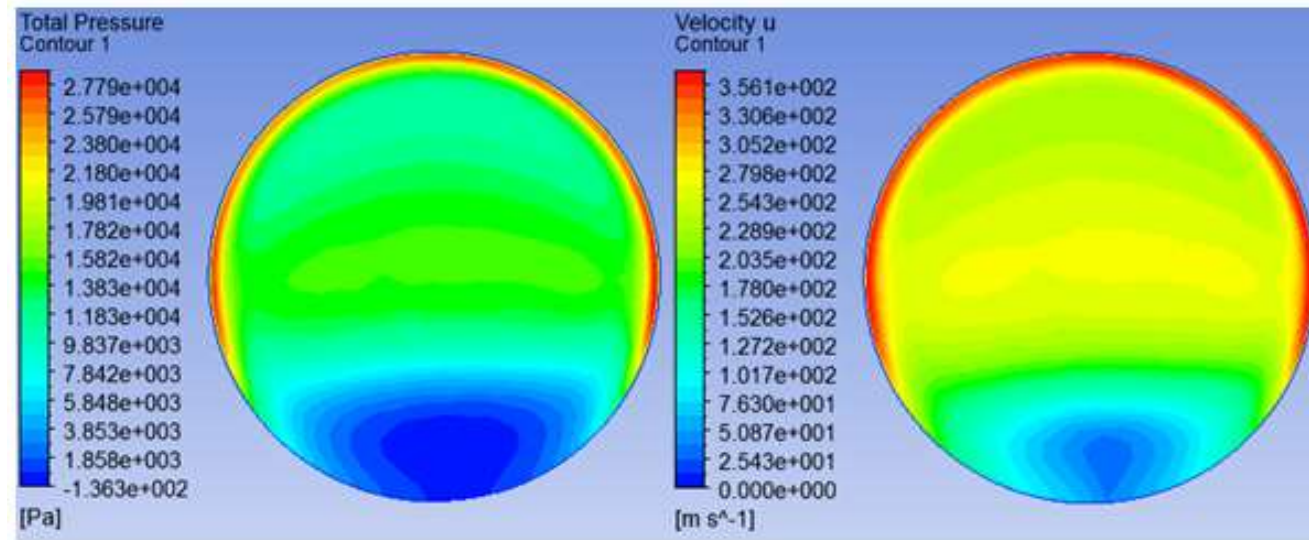
$L=7.27\text{m}$

Improved design



L	4.85m
H	1.615m
AR	1.75
CR	1.04
m/n	2
PR	98.04%
DC ₆₀	20.55%

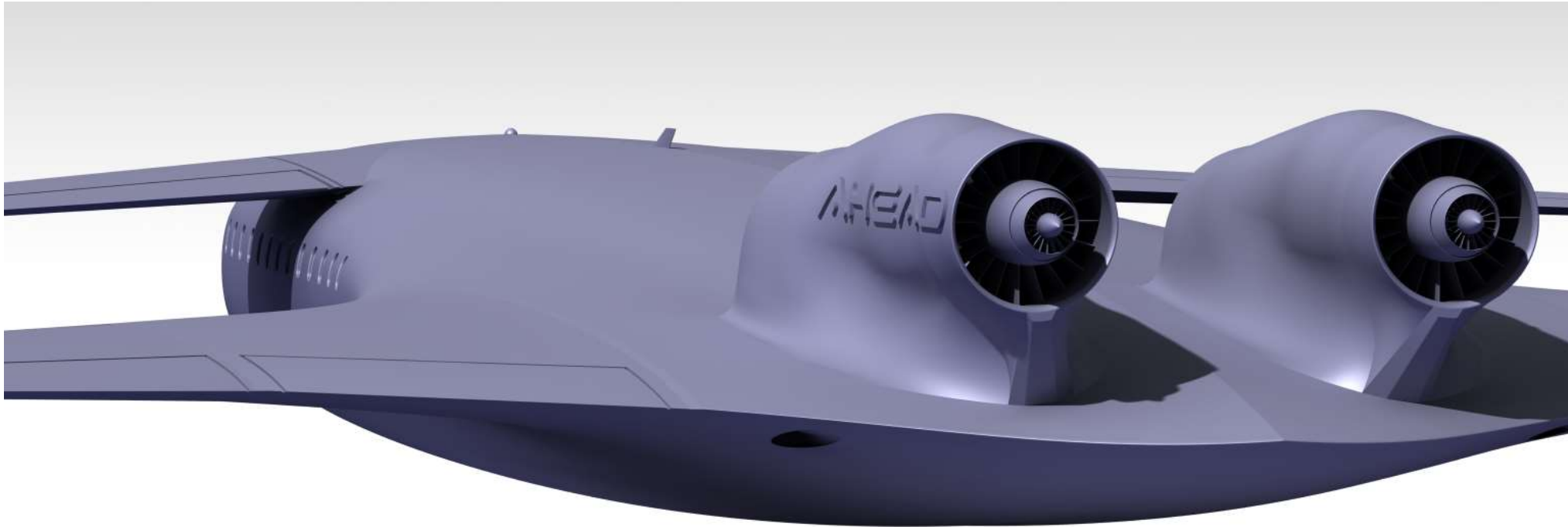
A. Gangoli Rao, A. Sharma, R. van Dijk,
"A CFD Based Parametric Analysis of S-
shaped Inlet for a Novel Blended Wing
Body Aircraft", 2017



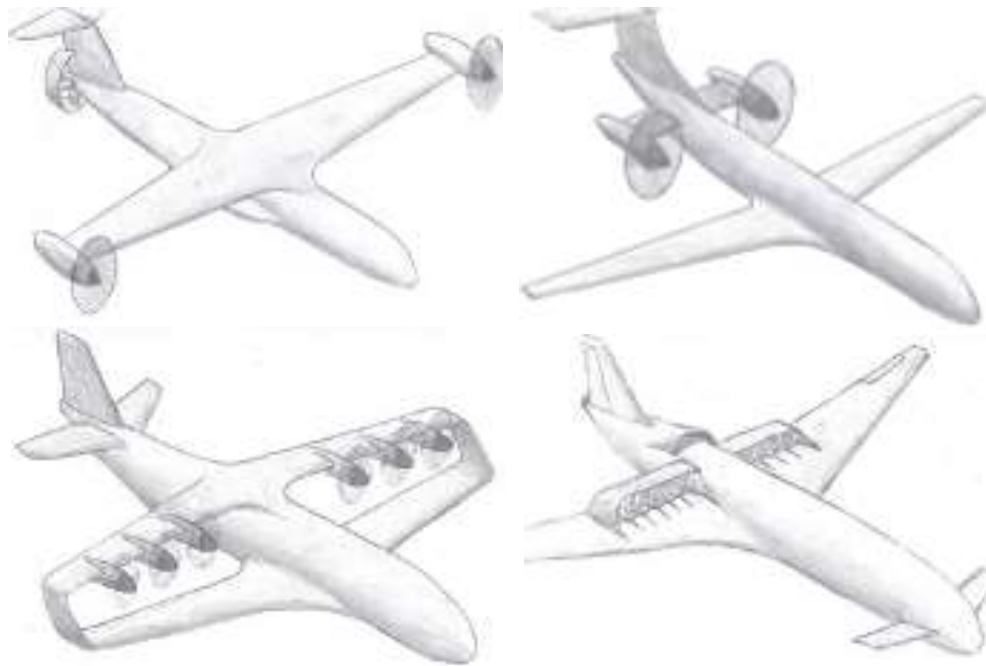
Total Pressure

Velocity

Revised design



Thank You



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